

PREOBRAZHENKA, Russian Federation

WMO: 249180

Lat: 60.050N

Lon: 108.083E

Elev: 968

StdP: 14.19

Time Zone: 8.00 (E08)

Period: 83-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -51.0	(c) -47.2	(d) -55.5	(e) 0.2	(f) -50.9	(g) -51.5	(h) 0.3	(i) -47.0	(j) 14.6	(k) 6.7	(l) 12.2	(m) -0.8	(n) 1.3	(o) 140	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.0	(c) 85.4	(d) 65.1	(e) 82.0	(f) 64.0	(g) 78.0	(h) 61.9	(i) 67.4	(j) 80.2	(k) 65.5	(l) 77.9	(m) 63.7	(n) 75.3	(o) 5.4	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	88.9	72.3	61.0	83.1	70.6	58.9	77.1	68.4	32.5	79.8	31.0	78.3	29.6	75.6	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.5	(b) 14.1	(c) 12.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			N/A	-54.0	N/A	71.6	N/A	5.4	N/A	-57.9	N/A	73.3	N/A	-61.1	N/A	74.8
			N/A	71.6	N/A	5.4	2.5	N/A	73.3	-61.1	N/A	74.8	N/A	-64.1	N/A	76.2
			N/A	71.6	N/A	5.4	2.5	N/A	73.3	-61.1	N/A	74.8	N/A	-64.1	N/A	76.2
			N/A	71.6	N/A	5.4	2.5	N/A	73.3	-61.1	N/A	74.8	N/A	-64.1	N/A	76.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.2	-16.1	-10.6	11.2	27.5	42.3	59.7	63.0	57.4	43.6	24.0	-1.1	-13.0	
	DBStd	31.06	17.01	17.29	14.83	11.76	8.50	7.54	5.67	6.60	8.42	10.91	17.31	18.58	
	HDD50	10437	2049	1696	1202	675	267	11	0	15	228	807	1534	1953	
	HDD65	14991	2514	2116	1667	1125	704	192	106	250	643	1272	1984	2418	
	CDD50	1014	0	0	0	0	28	302	405	244	35	0	0	0	
	CDD65	92	0	0	0	0	0	32	45	15	0	0	0	0	
	CDH74	1758	0	0	0	0	12	699	773	262	12	0	0	0	
	CDH80	486	0	0	0	0	0	194	223	69	0	0	0	0	
Wind	WSAvg	5.2	4.5	4.6	6.2	6.9	6.3	5.3	4.6	4.5	4.9	5.1	5.3	4.3	
Precipitation	PrecAvg	13.6	0.6	0.4	0.4	0.7	1.4	1.7	1.9	2.1	1.3	1.1	1.0	0.8	
	PrecMax	18.2	1.1	1.1	1.3	1.8	3.1	3.1	6.2	5.0	2.5	2.8	1.9	1.5	
	PrecMin	9.4	0.3	0.2	0.1	0.1	0.4	0.2	0.2	0.2	0.3	0.3	0.6	0.3	
	PrecStd	2.0	0.2	0.2	0.2	0.4	0.7	0.8	1.1	0.9	0.6	0.5	0.3	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.6	28.6	47.4	58.5	75.0	89.3	90.6	86.6	75.6	50.8	40.1	29.7	
		MCWB	22.8	26.4	37.2	45.0	54.3	65.3	66.4	66.0	59.8	41.4	37.6	28.0	
	2%	DB	17.7	25.8	40.9	52.8	68.5	85.6	85.4	81.0	68.5	45.2	32.9	23.0	
		MCWB	16.7	24.3	34.0	40.9	50.5	64.2	66.0	64.0	55.9	39.1	31.2	21.8	
	5%	DB	13.2	20.3	35.7	48.5	63.4	82.2	82.1	75.8	63.1	41.1	28.4	19.0	
		MCWB	12.3	18.7	30.4	38.2	48.6	62.7	65.0	62.1	53.0	36.9	26.7	17.9	
	10%	DB	9.1	16.0	31.3	44.4	58.5	78.1	78.1	71.6	57.7	37.5	24.1	14.8	
		MCWB	8.3	14.4	27.4	36.1	46.2	60.4	63.1	59.5	49.9	34.1	22.5	14.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	26.9	38.0	45.3	56.7	68.3	70.9	68.5	61.3	42.8	38.4	29.1
			MCDB	24.1	28.0	44.4	56.7	70.4	82.2	82.8	80.6	73.1	48.4	39.8	30.7
2%		WB	16.7	24.2	34.1	41.5	52.3	66.1	68.2	65.7	57.6	39.9	31.3	22.3	
		MCDB	17.6	26.2	40.1	50.5	64.7	81.8	81.4	76.5	66.2	44.4	32.8	23.1	
5%		WB	12.3	18.7	30.9	39.1	49.5	64.3	66.3	63.3	54.1	37.2	26.7	18.0	
		MCDB	13.1	20.3	35.4	47.5	62.2	78.6	77.8	72.9	60.9	41.2	28.3	18.9	
10%		WB	8.2	14.5	27.7	36.6	47.1	62.3	64.5	61.3	50.8	34.5	22.5	14.0	
		MCDB	9.1	16.0	31.6	43.7	57.4	75.1	75.9	69.7	56.1	37.1	24.2	14.7	
Mean Daily Temperature Range	5% DB	MDBR	17.6	24.2	26.0	22.3	20.4	24.8	23.0	22.3	19.0	13.5	16.7	16.1	
		MCDBR	19.7	17.3	24.3	24.6	28.7	33.4	31.4	30.7	29.3	17.6	15.9	14.9	
		MCWBR	19.4	15.8	18.4	15.6	15.0	14.8	14.5	15.3	17.0	11.9	15.4	14.6	
	5% WB	MCDBR	19.6	17.1	23.3	21.8	25.5	29.2	27.3	26.5	25.4	15.9	15.6	15.7	
		MCWBR	19.3	15.5	18.3	14.6	14.2	13.7	13.8	14.1	15.5	11.2	15.4	15.4	
Clear-Sky Solar Irradiance	taub	0.252	0.289	0.312	0.343	0.372	0.402	0.461	0.425	0.339	0.305	0.251	0.225		
	taud	1.962	2.010	2.050	2.112	2.295	2.314	2.132	2.197	2.435	2.301	2.009	1.905		
	Ebn at Noon	153	216	253	269	271	263	241	240	250	216	151	112		
	Edh at Noon	16	27	36	41	37	37	43	37	25	20	15	11		
All-Sky Solar Radiation	RadAvg	158	463	968	1497	1647	1834	1651	1301	798	370	189	88		
	RadStd	10	30	56	92	114	136	147	106	71	45	20	7		

Historical Trends

DBAvg	Heating		Cooling			Degree-Days			
	99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	+4.12	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page